

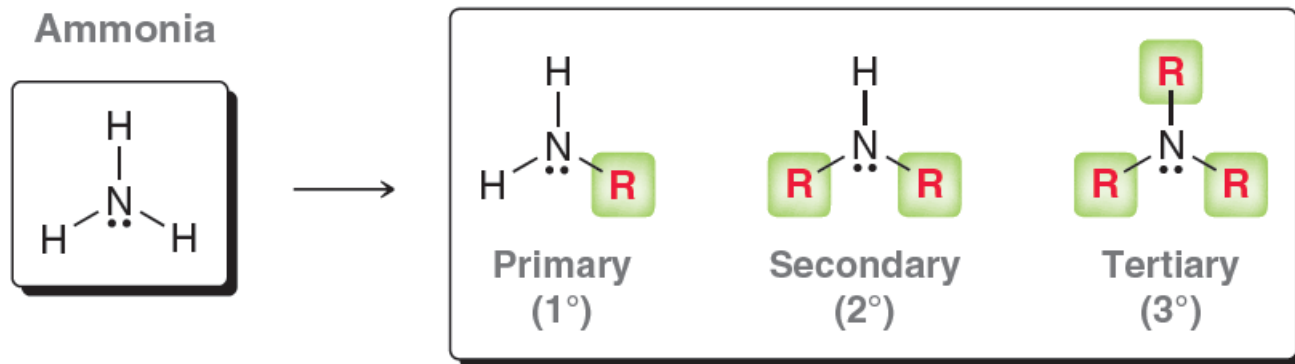
第十二章 胺

- 12.1 胺类化合物的命名及性质
- 12.2 胺类化合物的制备
 - 12.2.1 亲核取代
 - 12.2.2 叠氮法
 - 12.2.3 Gabriel合成法
 - 12.2.4 还原胺化
- 12.3 胺的酰基化
- 12.4 Hofmann消除

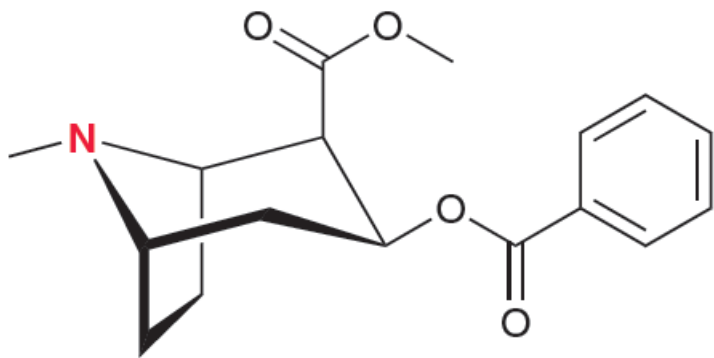
有机含氮物
(含N官能团)

12.1 胺类化合物

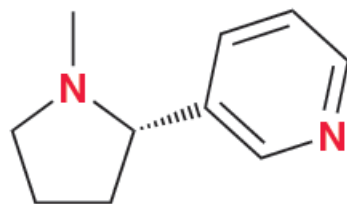
- 氨及其衍生物



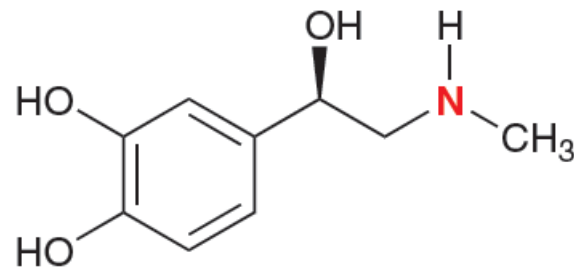
- 生物体内的胺类化合物



可卡因



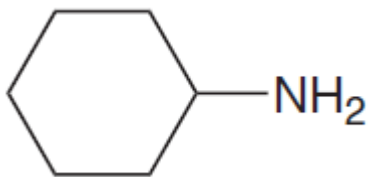
尼古丁



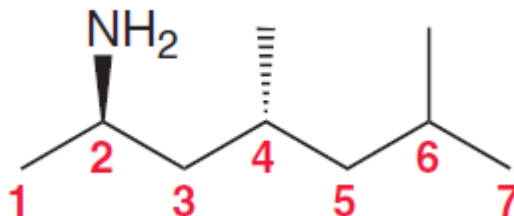
肾上腺素

一级胺的命名

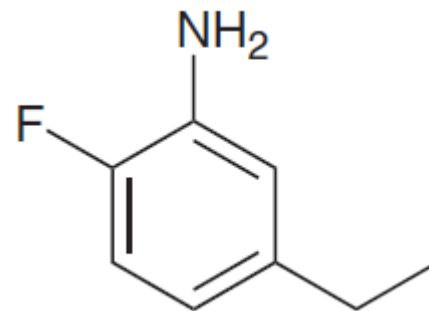
- 胺作为母体



环己基胺

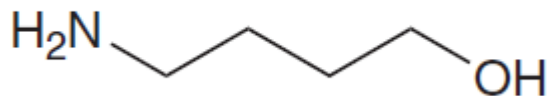


(2*R*,4*R*)-4,6-二甲基-2-庚胺

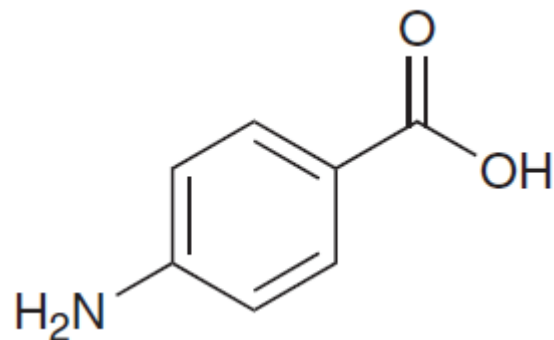


5-乙基-2-氟苯胺

- 胺作为取代基



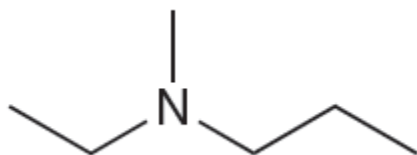
4-氨基丁醇



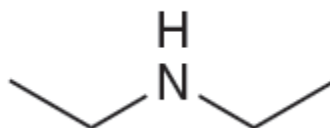
对氨基苯甲酸

二级、三级胺的命名

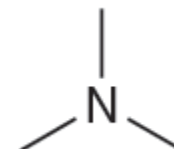
- 较简单的二级、三级胺



甲基乙基丙基胺

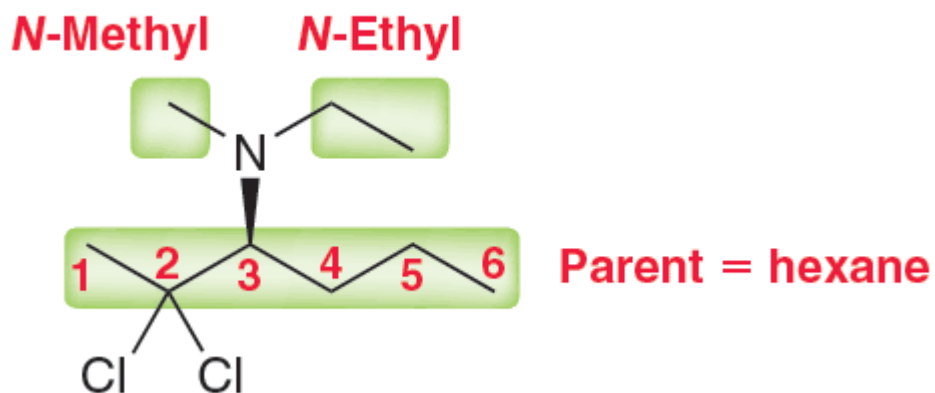


二乙胺



三甲胺

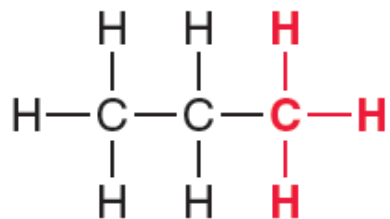
- 较复杂的二级、三级胺



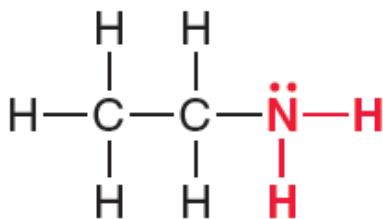
(S)-2,2-二氯-N-甲基-N-乙基-3-己胺

胺的性质

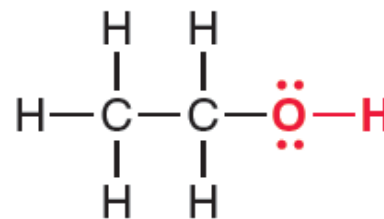
- 胺类化合物的沸点



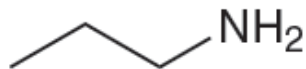
Propane
bp $\equiv -42^{\circ}\text{C}$



Ethylamine
bp $\equiv 17^{\circ}\text{C}$



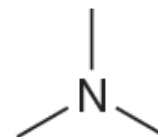
Ethanol
bp $\equiv 78^{\circ}\text{C}$



Propylamine
bp $= 50^{\circ}\text{C}$

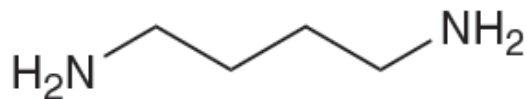


Ethylmethanamine
bp $= 34^{\circ}\text{C}$

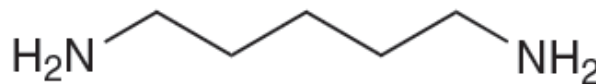


Trimethylamine
bp $= 3^{\circ}\text{C}$

- 胺类化合物有特殊气味



Putrescine
腐胺

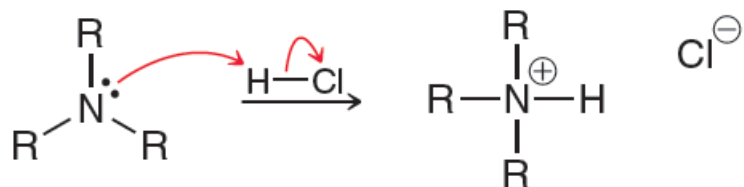


Cadaverine
尸胺

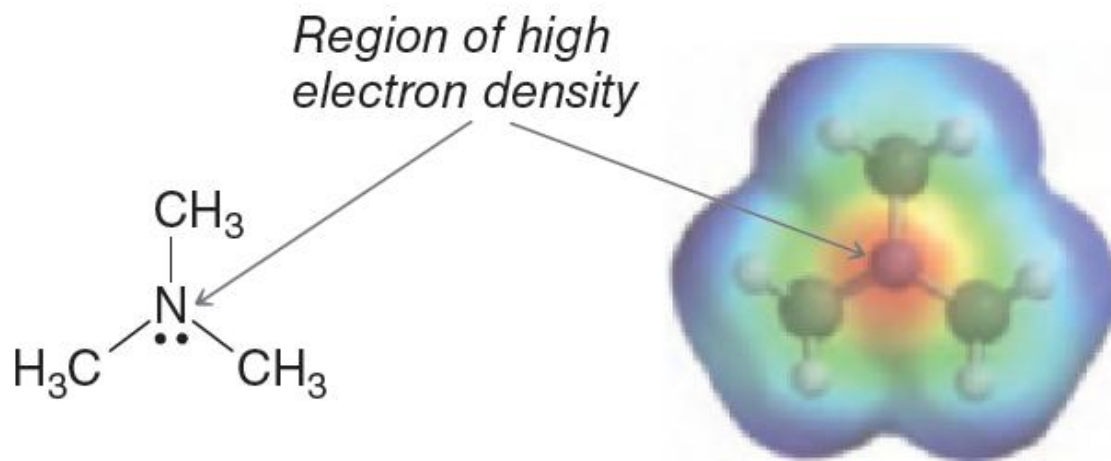
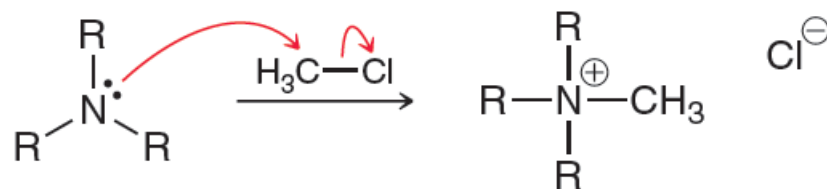
胺的反应性

- 胺主要体现氮原子的反应性

胺的碱性

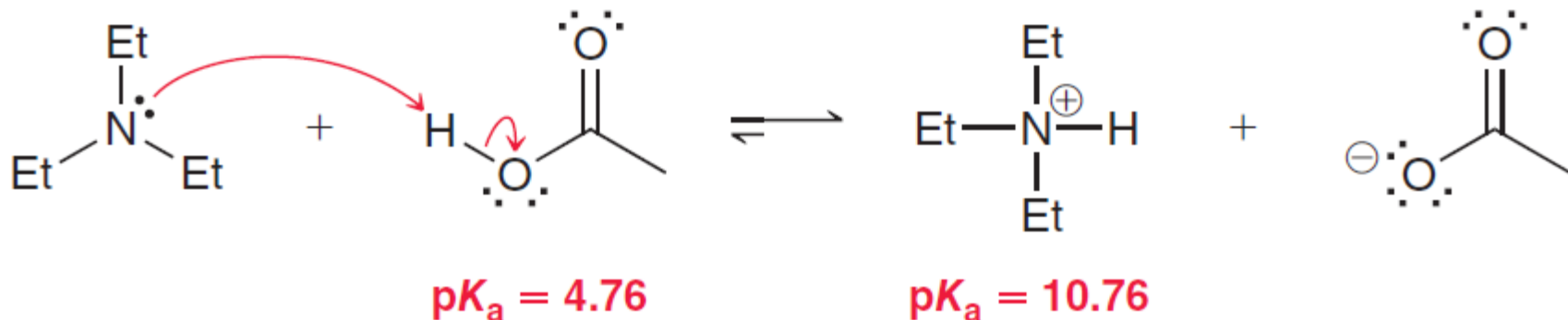


胺的亲核性

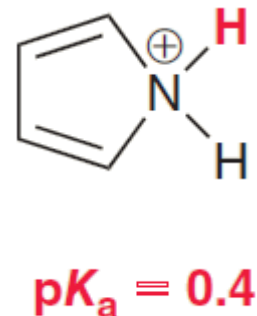
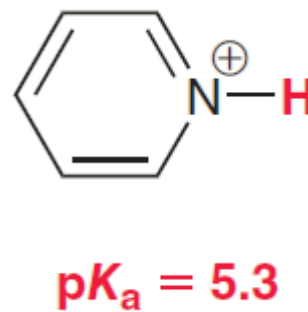
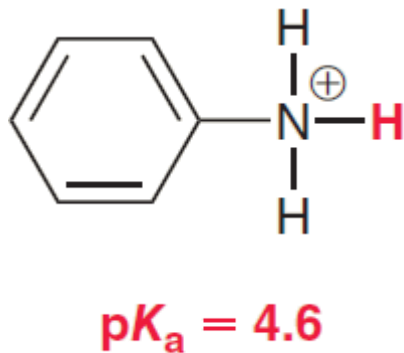
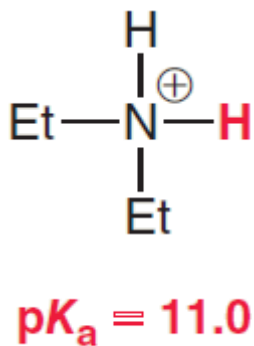
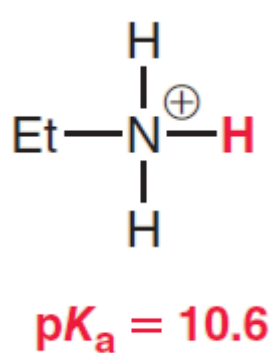


胺的碱性

- 用共轭酸的 pK_a 衡量胺的碱性

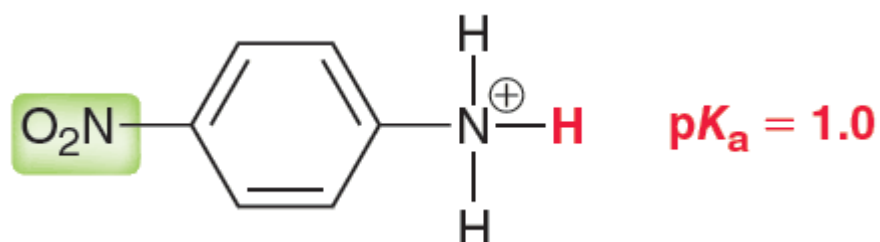
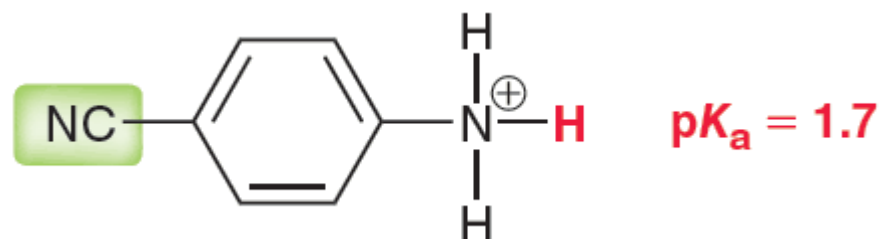
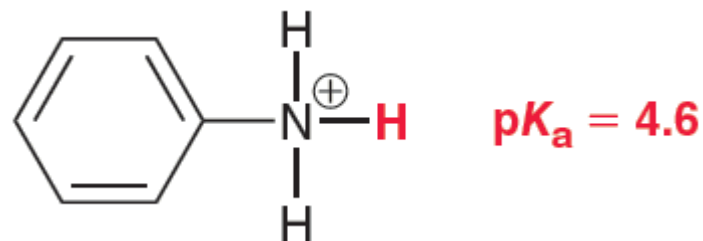
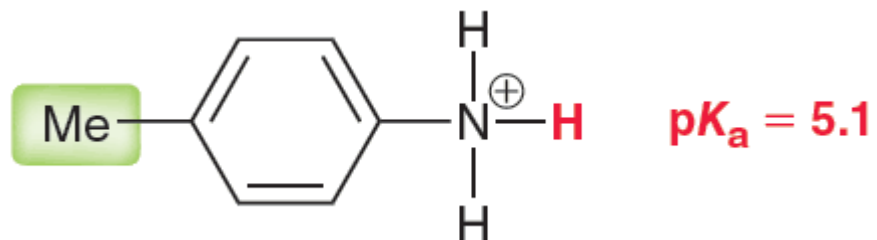


- 几类常见胺类化合物的碱性

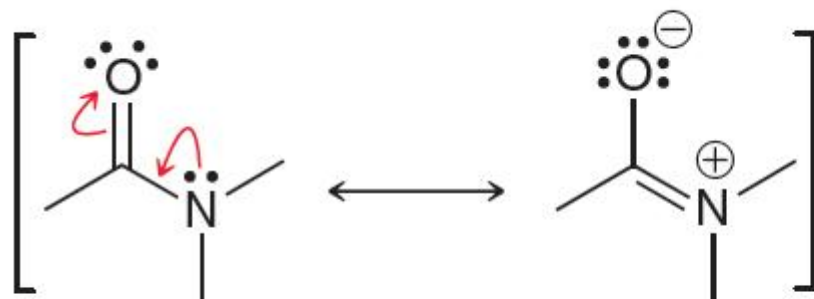


胺的碱性

- 苯胺类化合物的碱性

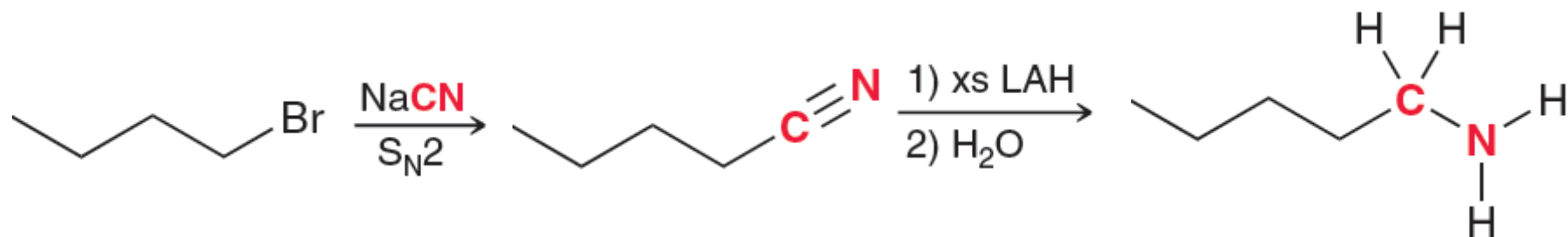


- 酰胺的碱性很弱

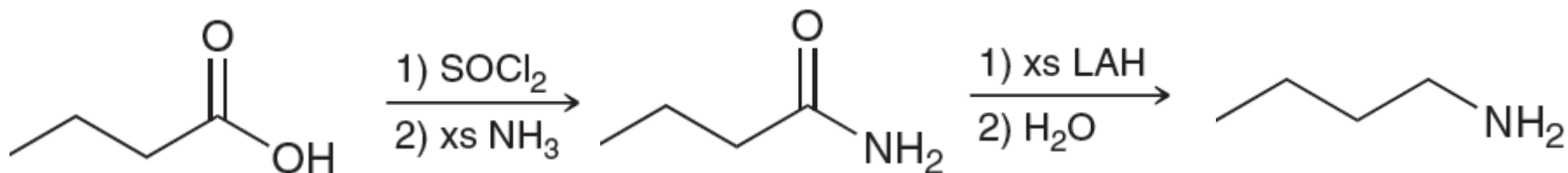


12.2 胺的制备

- 氰化物的还原制备一级胺

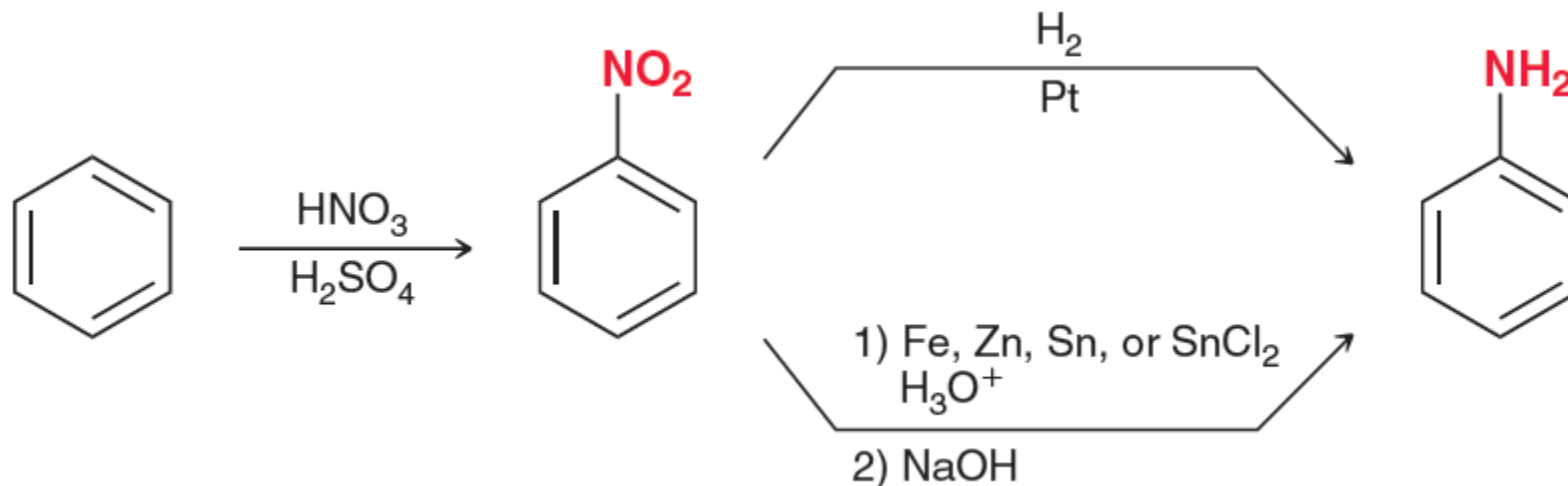


- 酰胺的还原制备一级胺

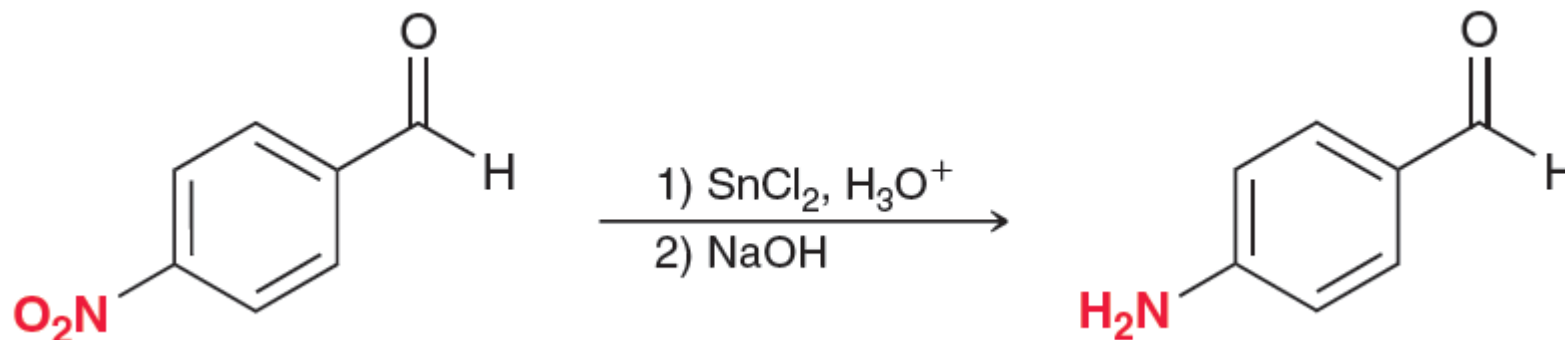


12.2.1 胺的制备

- 硝基苯的还原制备苯胺

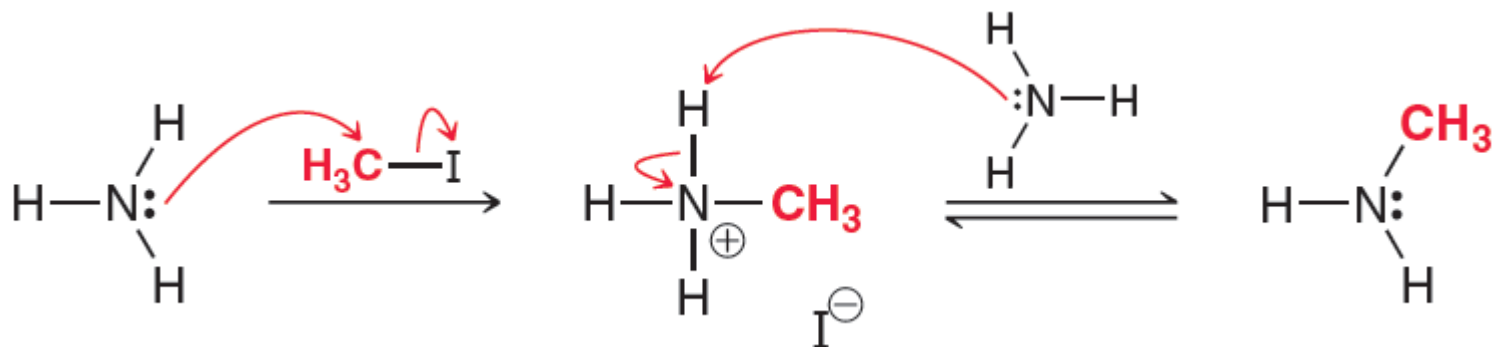


- 硝基苯还原的选择性

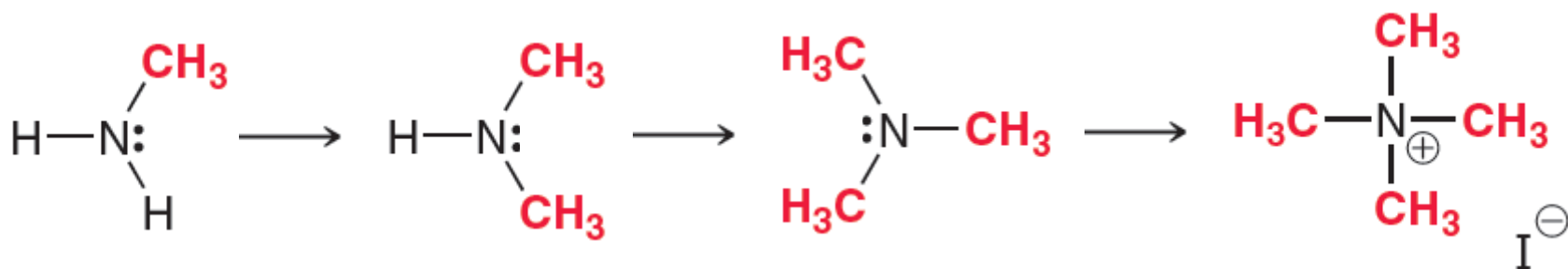


12.2.1 胺的制备

- 氨的亲核取代制备一级胺



- 胺的进一步亲核取代



一级胺

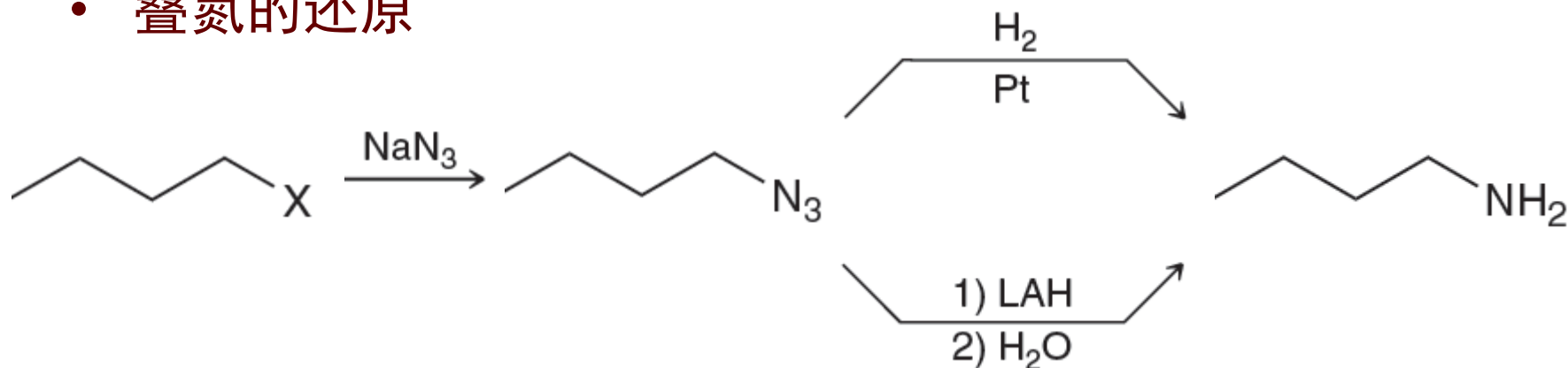
二级胺

三级胺

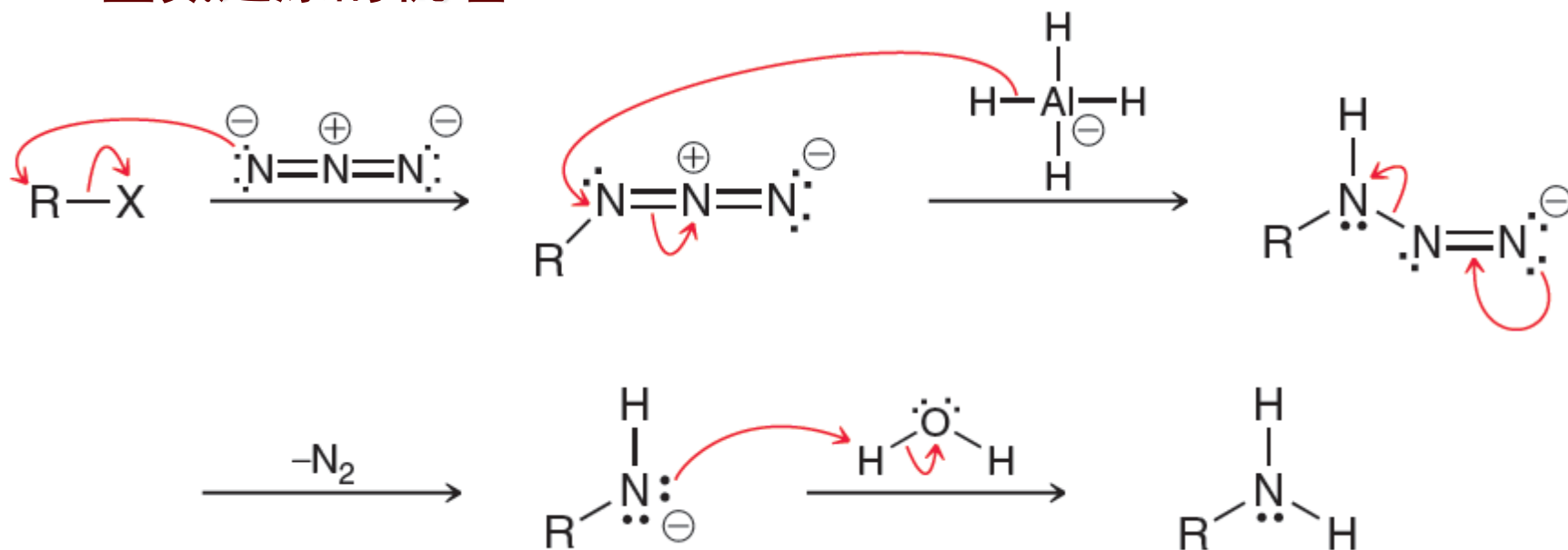
四级铵盐

12.2.2 叠氮法合成一级胺

- 叠氮的还原

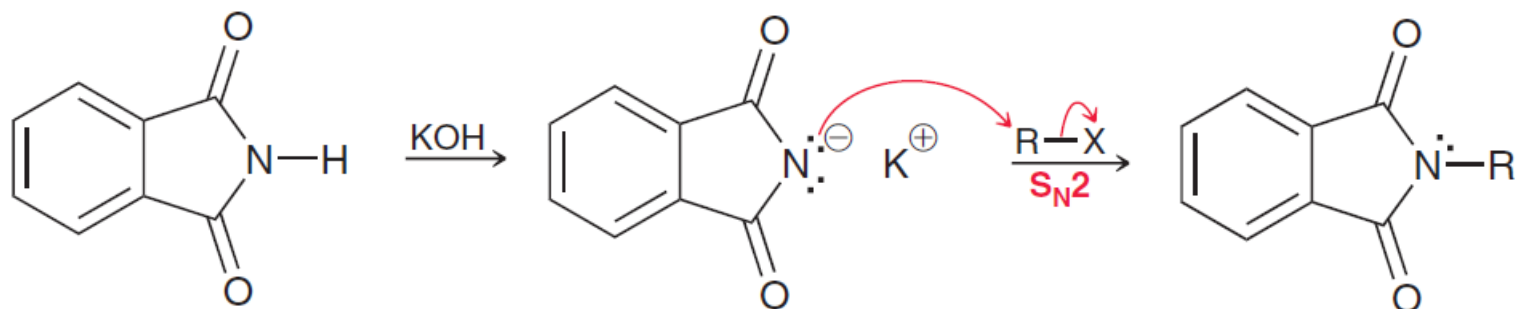


- 叠氮还原的机理

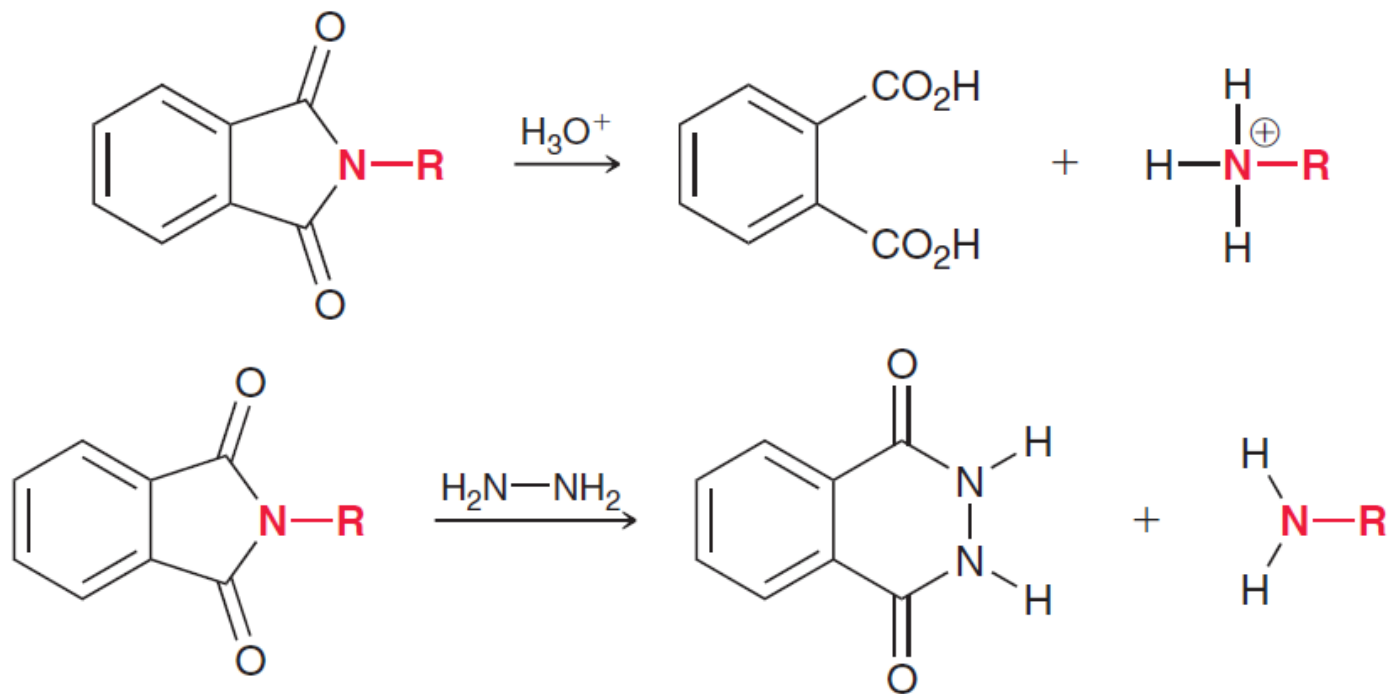


12.2.3 Gabriel合成法

- 邻苯二甲酰亚胺的亲核取代



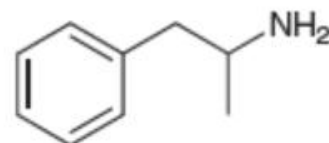
- 邻苯二甲酰亚胺的水解



LEARN the skill

Amphetamine is a stimulant (stronger than caffeine) that increases alertness and decreases appetite. It was used extensively during World War II to prevent combat fatigue.

练习:



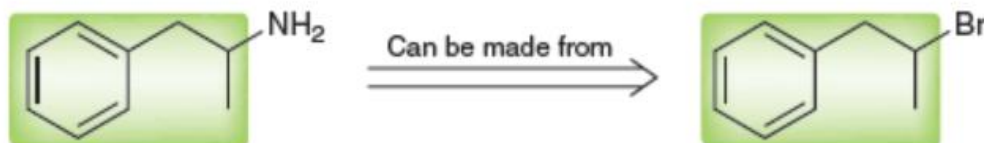
Amphetamine

Propose a method for preparing amphetamine that utilizes a Gabriel synthesis.



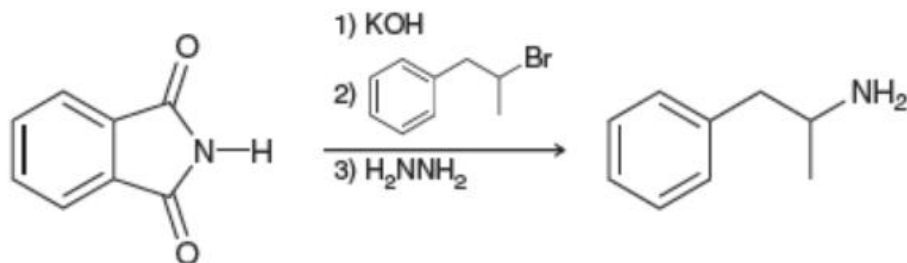
SOLUTION

First identify an alkyl halide that can serve as a precursor in the preparation of amphetamine. To draw this precursor, simply replace the amino group with a halide:



STEP 1
Identify the appropriate alkyl halide to use.

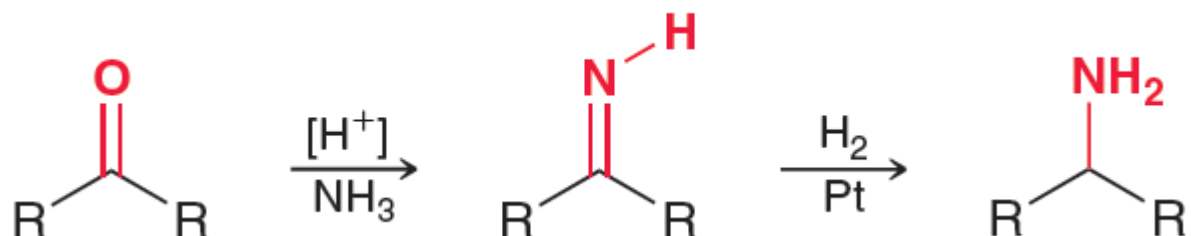
In the Gabriel synthesis, the nitrogen atom comes from phthalimide, which serves as the starting material. The process requires three steps: (1) deprotonation to form potassium phthalimide, (2) nucleophilic attack with an alkyl halide, and (3) releasing the amine via hydrolysis or via treatment with hydrazine. Draw the reagents for all three steps of the Gabriel synthesis:



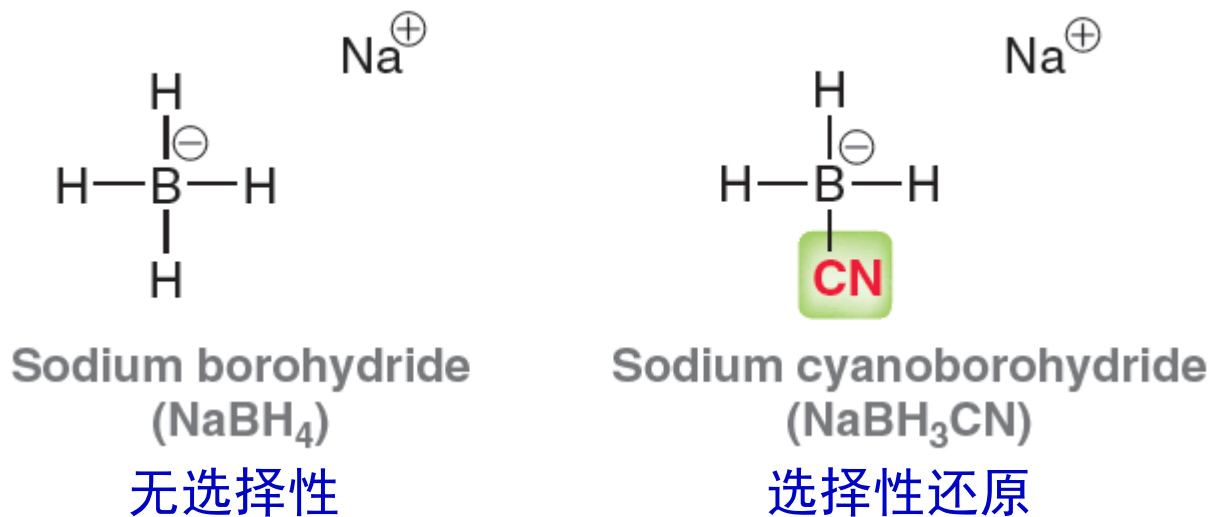
STEP 2
Draw all three steps, starting with phthalimide.

12.2.4 还原胺化

- 原位生成的亚胺的还原

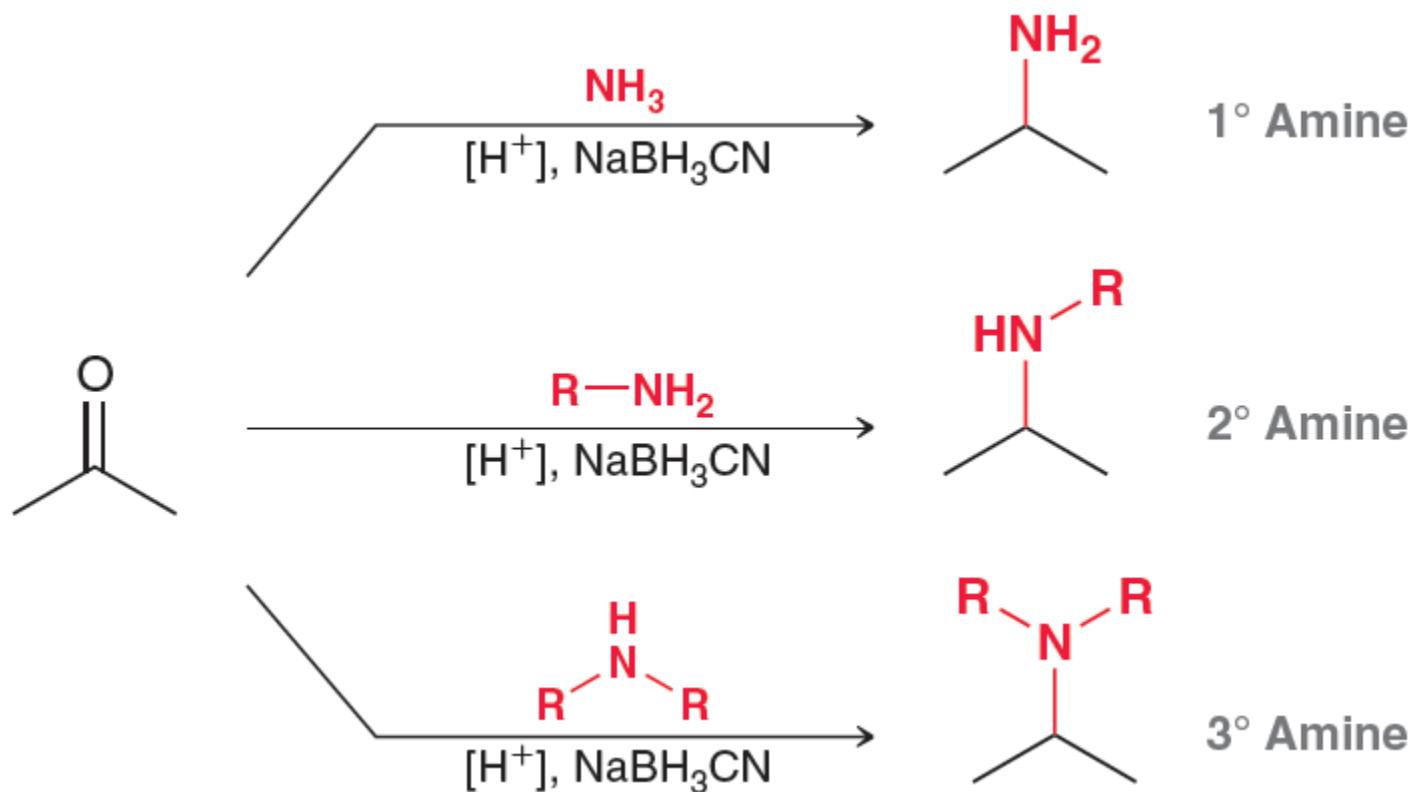


- 氰基硼氢化钠作为还原剂



12.2.4 还原胺化

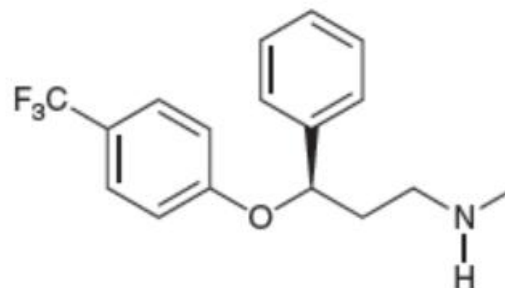
- 还原胺化用于制备一级、二级、三级胺



LEARN the skill

练习:

Fluoxetine, sold under the trade name Prozac, is used in the treatment of depression and obsessive compulsive disorder. Show two different ways of preparing fluoxetine via reductive amination.



Fluoxetine

STEP 1

Identify all C—N bonds.

STEP 2

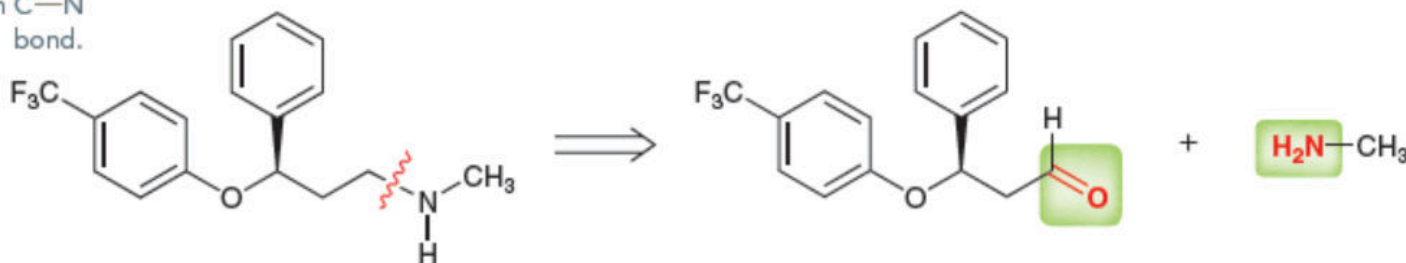
Apply a retrosynthetic analysis of each C—N bond.



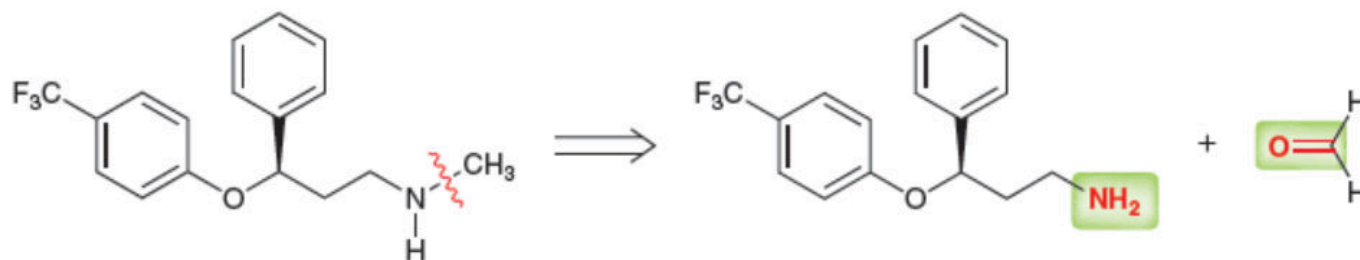
SOLUTION

Begin by identifying all C—N bonds. Then, using a retrosynthetic analysis, identify the starting amine and the carbonyl compound that could be used to make each of the C—N bonds:

Method 1

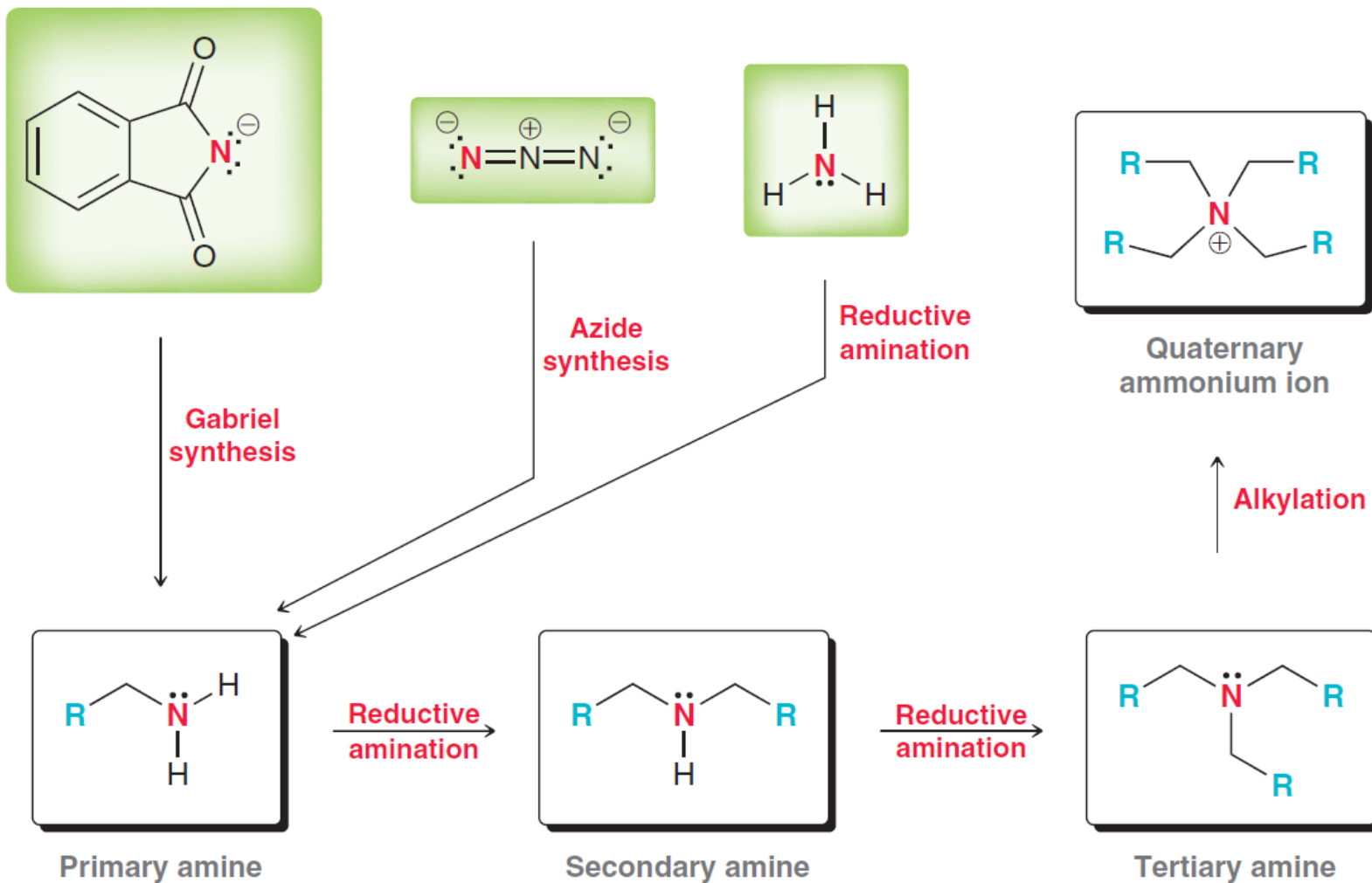


Method 2



胺的制备小结

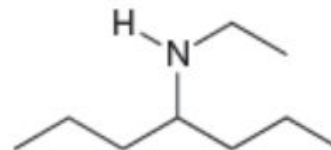
- 从简单原料出发合成各类胺类化合物



LEARN the skill

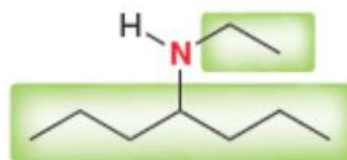
练习:

Using ammonia as your source of nitrogen, show what reagents you would use to prepare the following amine:

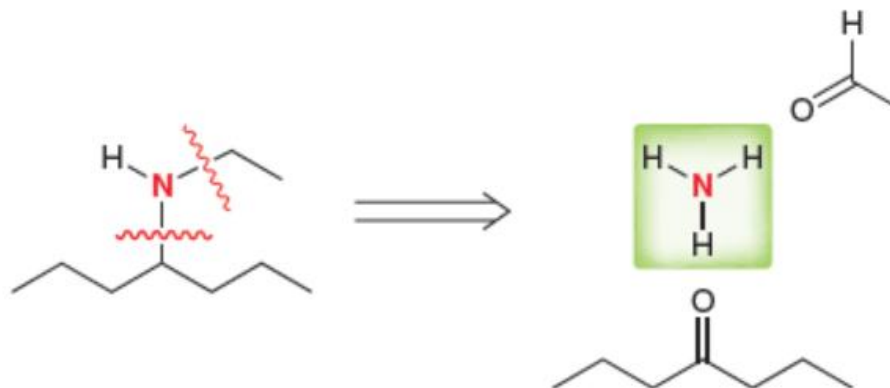


SOLUTION

Begin by identifying the alkyl groups connected to the nitrogen atom and determine whether the amine is primary, secondary, or tertiary.



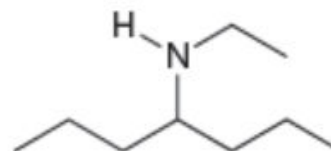
In this case, the amine is secondary because the nitrogen atom bears two alkyl groups and one hydrogen atom. Each of the highlighted groups must be installed separately. The source of nitrogen is ammonia, which dictates that both groups must be installed via reductive amination. The following retrosynthetic analysis reveals the necessary starting materials:



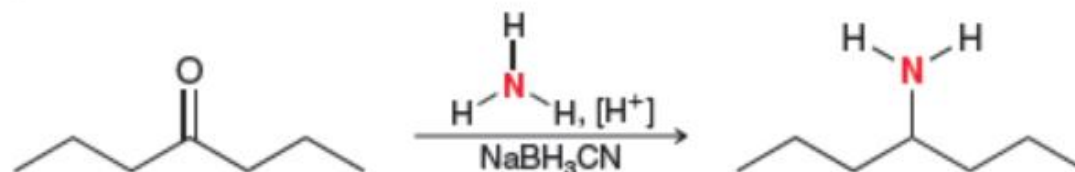
LEARN the skill

练习:

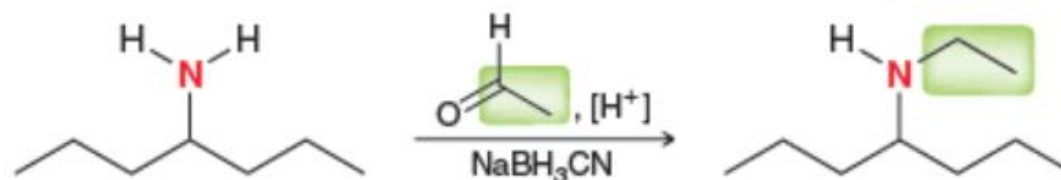
Using ammonia as your source of nitrogen, show what reagents you would use to prepare the following amine:



Each C—N bond in the product (left) can be formed via the reaction between ammonia and a ketone or aldehyde. This synthesis therefore requires two successive reductive aminations, which can be performed in either order. One reaction is used to install the first group,

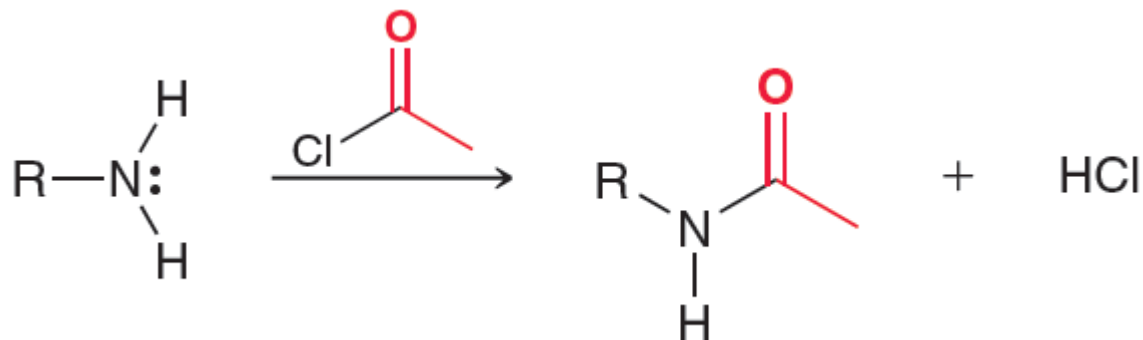


and another reductive amination is used to install the second group,

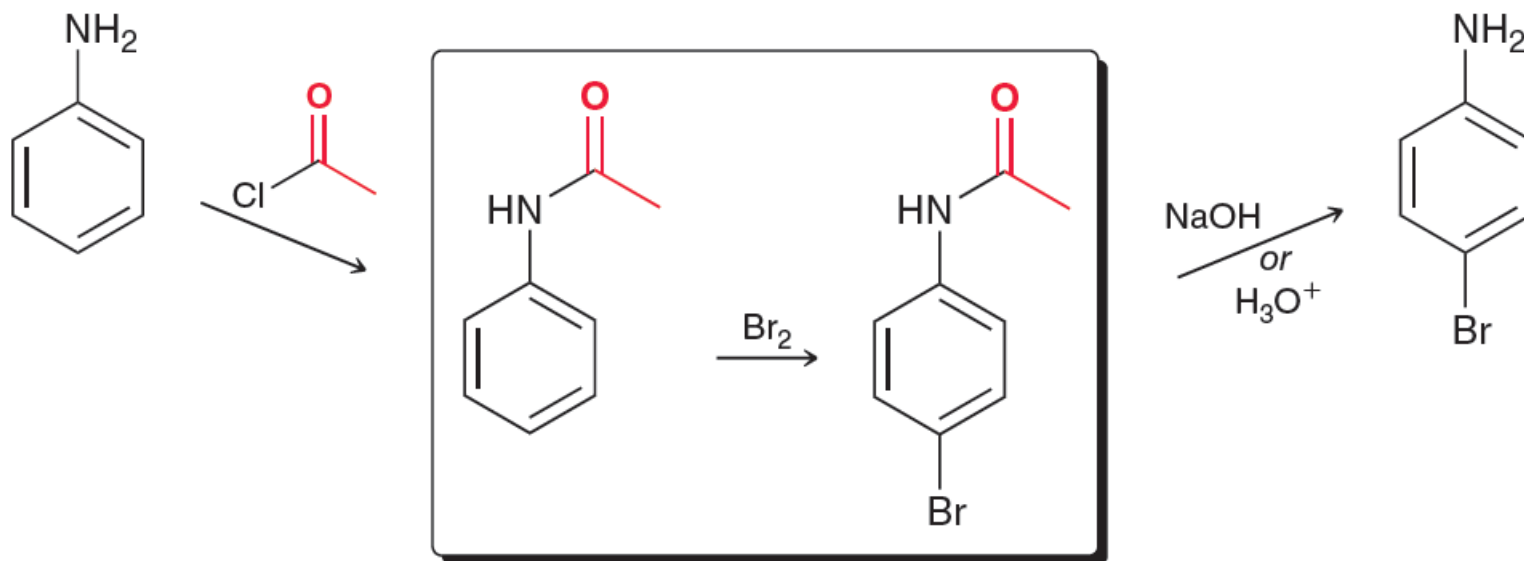


12.3 胺的酰基化

- 胺的酰化制备酰胺

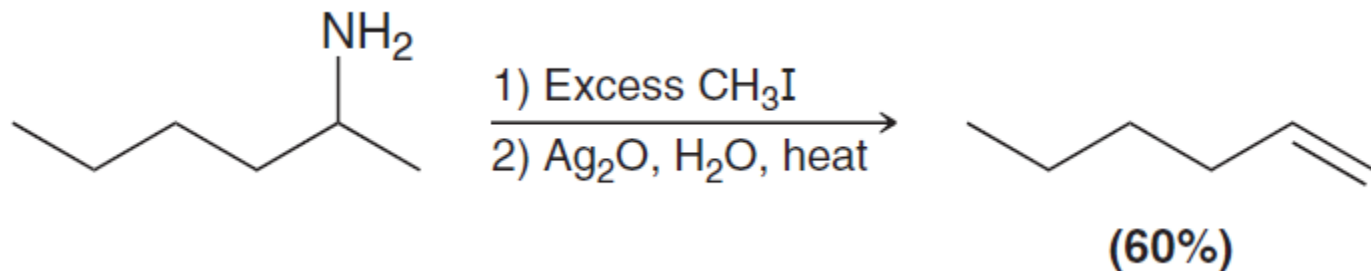


- 酰基作为保护基

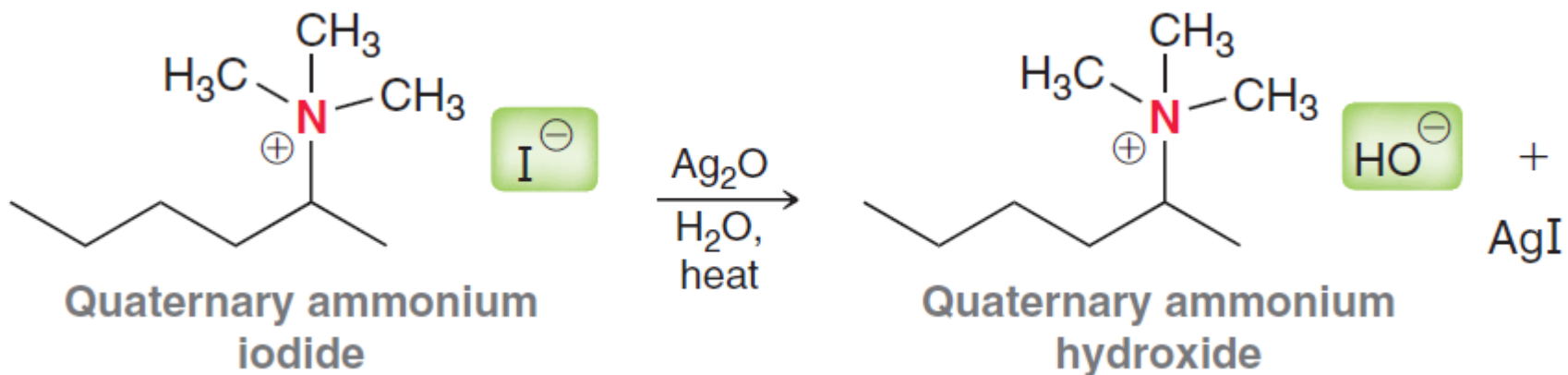


12.4 Hofmann消除

- 四级铵盐在碱性条件下的消除

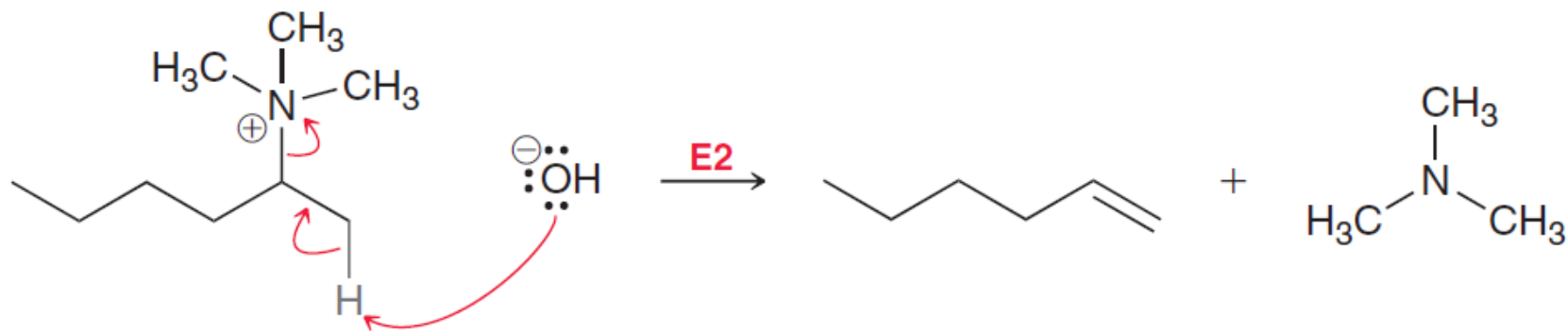


- Hofmann消除的条件

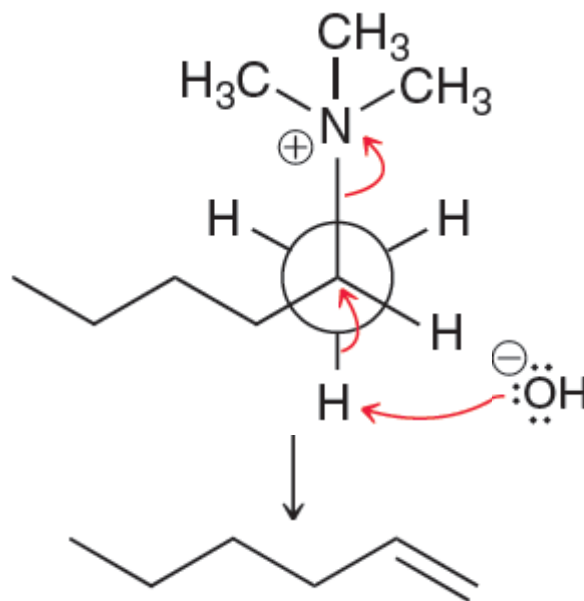
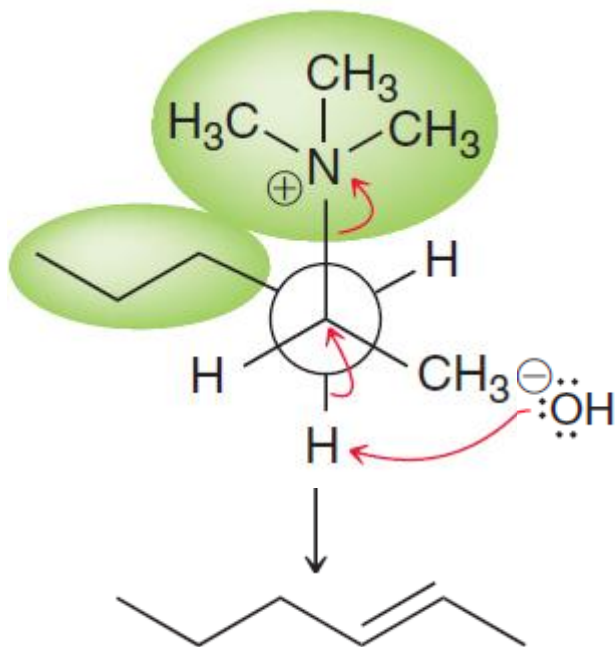


Hofmann消除的区域选择性

- Hofmann消除生成少取代烯烃



- Hofmann消除区域选择性：动力学控制



关于考试

期末考试：

6月8日（星期日）上午9:00-11:00

生44+生46：共33人，六教6A117；

其余班级：共35人，六教6A116。

关于考试

课后作业 15%

期中考试 35%

期末考试 50%

期末考试题型

一、判断下列结构的构型（4分）：（每题2分，共4分）

二、完成反应，如有立体化学请标明（每空1.5分，共36分）

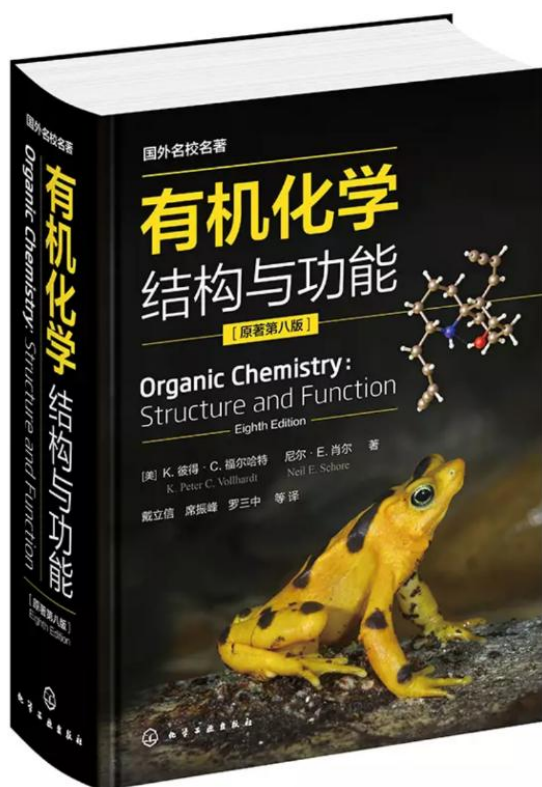
三、用箭头画出以下转化的反应机理（以4题中得分最高的3题计入总分，共30分）

四、合成路线设计：（以4题中得分最高的3题计入总分，共30分）

5个平行教学班的期末考试试卷有70%题目相同，30%不同，期末占50%权重
重点考察从卤代烃到胺的内容

期中之前的内容不会重点考察，但不可能完全不涉及

期末将侧重运用知识的能力



奖励

课程总成绩 and
期末考试卷面成绩排名前5的同学

- 谢谢同学们的努力学习和付出
- 希望大家学有收获，学有所用
- 预祝大家期末考试顺利，取得好成绩
- 6月7日(明天)下午14:00-15:00 三教3200

有机化学专题讲座（合成策略）

本学期的课程到此结束！